Week 10 - midterm-test

Target: >=35

Chap 1

1. Software stack (???) for hosting ASP.NET web applications. // Page 3 - 4

- Operating System: Windows, Linux, or MacOS (supported by .NET 10)
- Web Server: Internet Information Services (IIS)
- Database: Any compatible database server
- Programming Runtime: .NET 10 (cross-platform)

2. Advantages of using the ASP.NET platform for developing web applications // Page 8-9

Existing software environment
- Many companies may already use Windows-based software

Customer support
- Windows-based software are usually paid

Expandability and scalability
- Companies and businesses will grow and expand

Consistent software components

3. The role of the 3 core components of MVC web architecture. // Page 27

	Model	View	Controller
	Data layer	Presentation layer	Application logic layer
	Represent data of the application	Template files the application users	Handle incoming requests
E.g.	Database	HTML, CSS	C# controller classes

4. Advantages of MVC web architecture // Page 28 - 31

Separation of concerns
- Each MVC components focuses on specify tasks (modular)

Task distribution
- Making it easier to distribute tasks among developers, database administrators and designers
  - Developers: focus on controllers
  - Database administrators: focus on models
  - Designers: focus on views

Reusable
- MVC application is modular

Unit testable
- Application logic (controller) is decoupled from data and presentation

5. Why URL routing is necessary for ASP.NET MVC web applications. // Page 50-51

-Web pages are not physical files, but result of interaction between model, view and controller
- URL routing engine maps incoming requests (URLs) against the routing rules

6. URL Mapping Analysis // Page 52-54 - read slides

Following the default routing pattern {controller}/{action}/{id}

Example Question 1- Discuss in words how the following URL is mapped: https://www.abc.com/Product/Update/P001?quantity=123
Ans:
Controller: Product.
Action Method: Update.
ID Parameter: P001.
Query String Parameter: quantity (value is 123)

Example Question 2 - Write the URL to access the StudentController and Detail action method. Pass S001 for id and 2025 for year.
Ans:
The URL would be: https://www.abc.com/Student/Detail/S001?year=2025

Code Format

```
public class _____ : Controller
{
  public IActionResult _____ ( _____ , _____ )
  {
    ...
  }
}
```

Example Question 3 - Rewrite the following code filling in the missing parts to map to:
https://www.abc.com/Product/Update/P001?quantity=123
Ans:

```
public class ProductController : Controller
{
        public IActionResult Update (string id, int quantity)
        {
        }
}
```

Chap 2

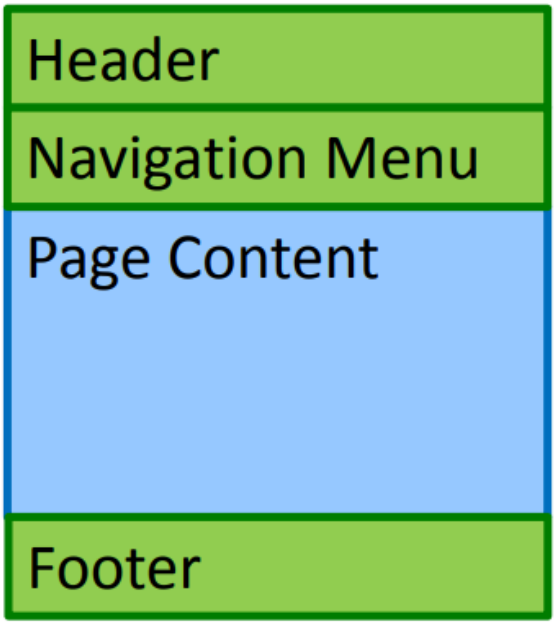
Suggest: all read with slides

1. Advantages of using reusable layout in ASP.NET MVC web applications? // Page 3-4

- Provide users with consistent experience as they navigate from page to page.
- Layout reduces duplicate code in views.

For example:

- Header
- Navigation Menu
- Footer



2. What are tag helpers? // Page 32

- Tag helpers extend the functionality of HTML elements by supporting non-standard custom HTML attributes.

3. Differentiate between tag helpers and HTML helpers. // Page 39 / Code part Page 38

Tag helpers	HTML helpers
Written as HTML elements with custom non-standard attributes	Written as C# methods. Work like function calls with parameters
Codes are clean, concise and easier to understand	Codes are usually more verbose and slightly complicated
Codes are more readable as they resemble typical HTML syntax	Codes are less readable as HTML and C# codes are mixed together

4. What are partial views? // Page 41

- is a reusable UI component used to render a portion of a web page.
- Commonly used to break down large views into smaller, manageable and reusable parts.

Chap 3

1. JavaScript (client-side processing) VS C# (server-side processing). // Page 3-4

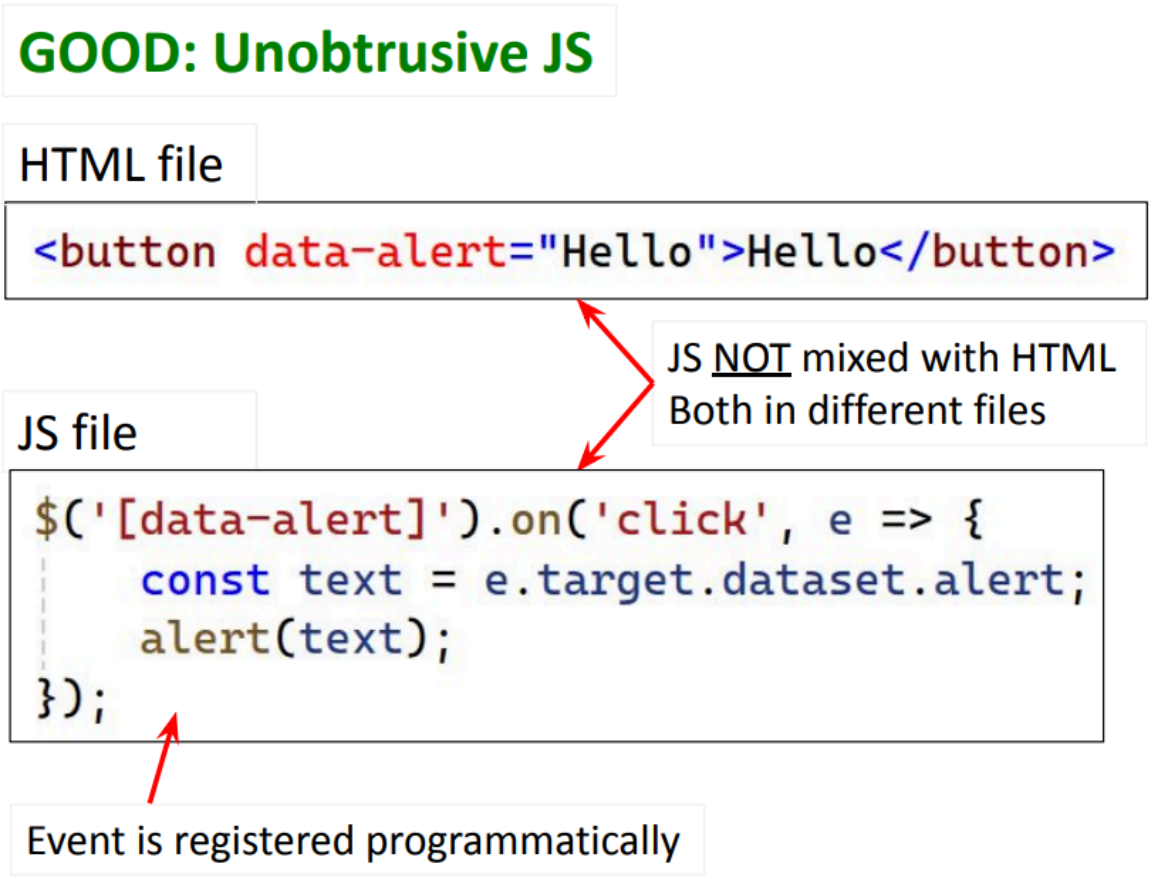
JavaScript (client-side processing)	C# (server-side processing)
runs locally on the client's browser	used for server-side processing
Less powerful	more powerful
No direct access to database	can directly access databases

2. Define unobtrusive JavaScript // Page 5

- Separation of JavaScript (behavior) from HTML (markup)
- Avoid the use of inline JavaScript for event handling



Unobtrusive JavaScript ...



3. Advantages of writing unobtrusive JavaScript. // Page 8 - 10

Modularity

- Making JS modules plug-able and reusable

Task distribution

- Web designers can work on HTML and CSS without worrying how to program JS code

Maintainability

- Also making the code easier to read (less messy)

4. Discuss the problems of the traditional page loading approach without AJAX. // Page 19

Traditional loading causes a full-page reload every time when a user clicks a link or submits a form. This results in a loss of view context

5. AJAX programming technique and its advantages. // Page 20

AJAX = Asynchronous JavaScript and XML
- Handle HTTP request and response using JavaScript running in the background
- Allow form submission and content update without the need of reloading whole page

Advantages: Improve responsiveness and user experience

6. Examples of AJAX features // Page 21

• AJAX searching
• AJAX sorting
• AJAX paging

7. AJAX polling technique // Page 50

- The process of continuously sending AJAX requests to server in the background to obtain updated data or content.
- AJAX requests are sent automatically, usually by a fixed interval , without user interaction.

8. Purpose of extension methods in C# // Page 57

- An extension method allows us to add new methods to existing types without modifying the original types.
- Helpful for adding functionality to classes where we do not have access to the source code.